

- 1.28 understand that the upward forces on a light beam, supported at its ends, vary with the position of a heavy object placed on the beam**
- 1.29 describe experiments to investigate how extension varies with applied force for helical springs, metal wires and rubber bands
- 1.30 understand that the initial linear region of a force-extension graph is associated with Hooke's law
- 1.31 describe elastic behaviour as the ability of a material to recover its original shape after the forces causing deformation have been removed.

d) Astronomy

Students will be assessed on their ability to:

- 1.32 understand gravitational field strength, g , and recall that it is different on other planets and the moon from that on the Earth
- 1.33 explain that gravitational force:
- causes moons to orbit planets
 - causes the planets to orbit the sun
 - causes artificial satellites to orbit the Earth
 - causes comets to orbit the sun
- 1.34 describe the differences in the orbits of comets, moons and planets
- 1.35 use the relationship between orbital speed, orbital radius and time period:

$$\text{orbital speed} = \frac{2 \times \pi \times \text{orbital radius}}{\text{time period}}$$

$$v = \frac{2 \times \pi \times r}{T}$$

- 1.36 understand that:
- the universe is a large collection of billions of galaxies
 - a galaxy is a large collection of billions of stars
 - our solar system is in the Milky Way galaxy.